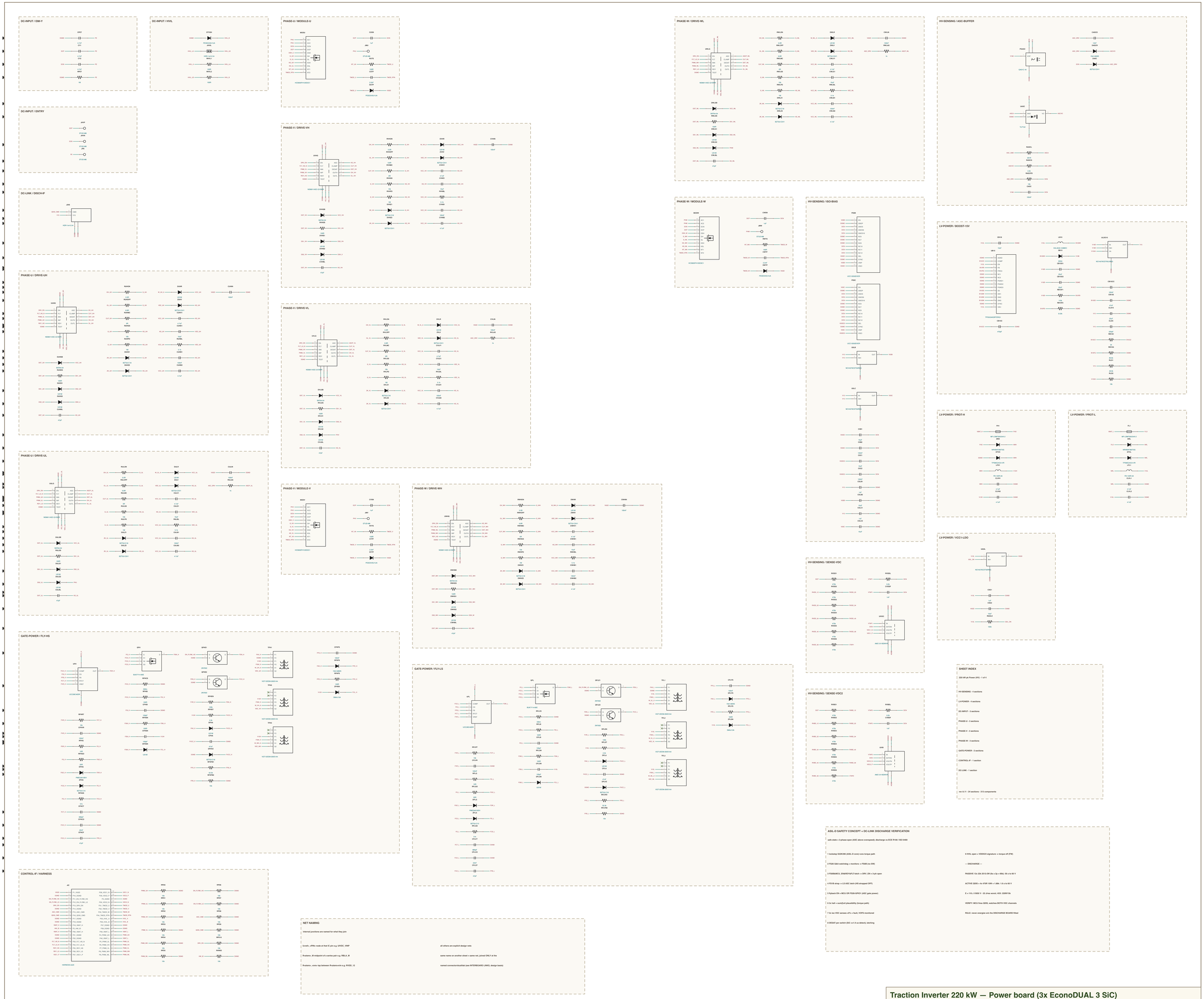

Traction Inverter 220 kW — Power board (3x EconoDUAL 3 SiC)

Traction Inverter 220 kW pk - Power (HV), sheet 1 of 4

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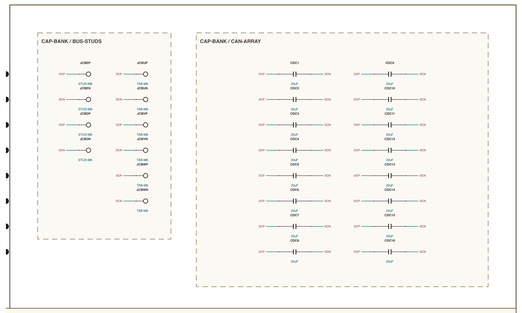

rter — Cap bank (busbar assembly, 320 uF)

20 kW pk - Cap bank (HV, busbar), sheet 2 of 4

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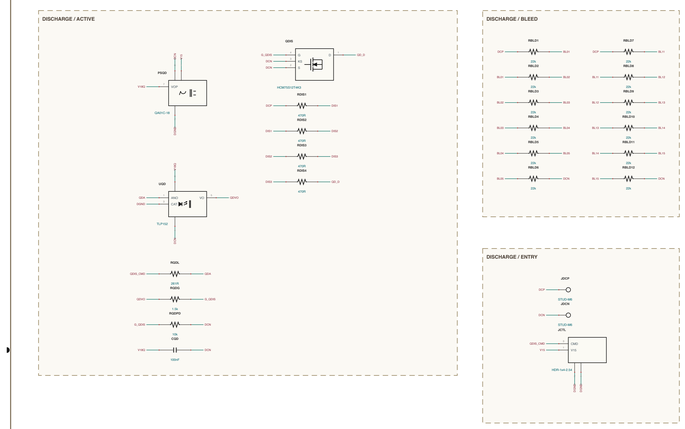

220 kW pk Discharge (HV, Bol

NET NAMING: Grands, uPRes o sma

Traction Inverter — Discharge board (bolt-on, XM3 pattern)

Traction Inverter 220 kW pk - Discharge (HV, bolt-on), sheet 3 of 4

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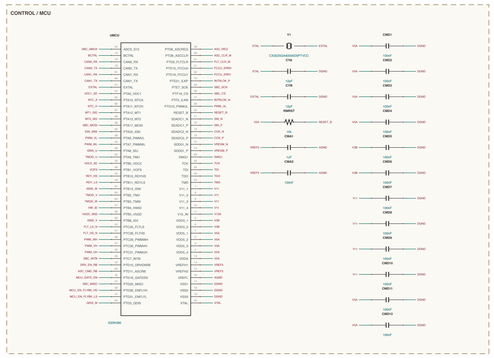
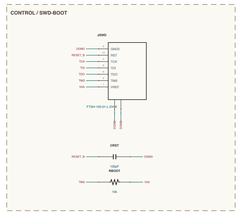
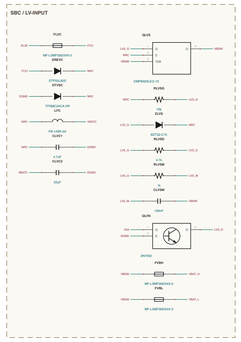
CONTROL / MCU



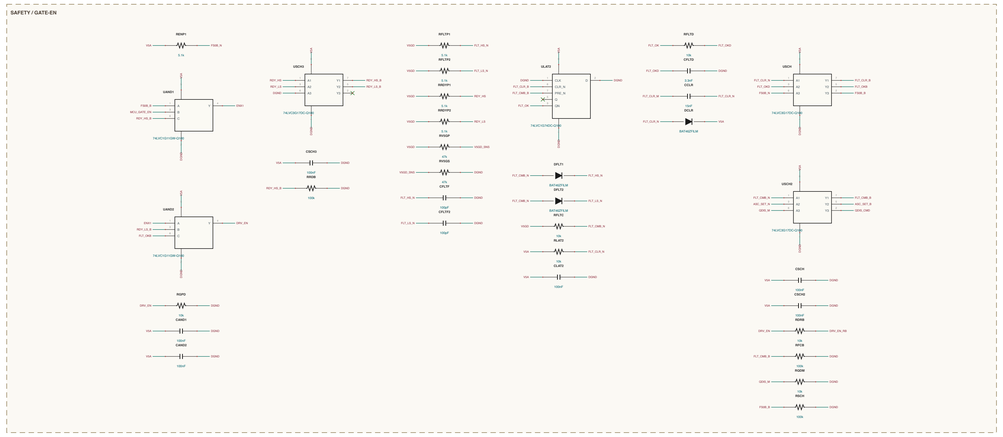
CONTROL / SWD-BOOT



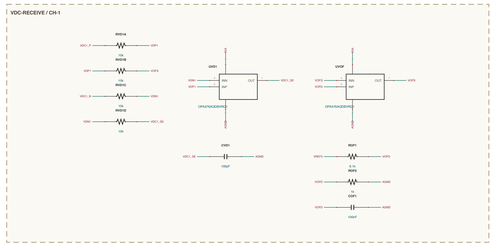
SBC / LV-INPUT



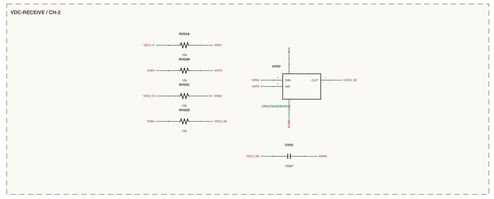
SAFETY / GATE-EN



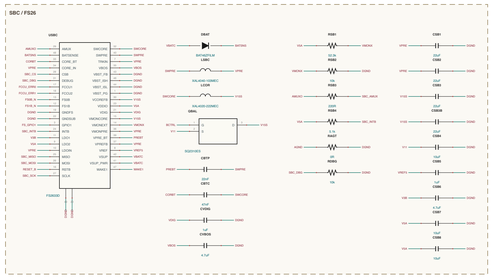
VDC-RECEIVE / CH-1



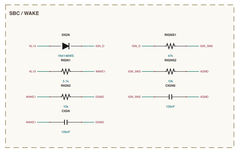
VDC-RECEIVE / CH-2



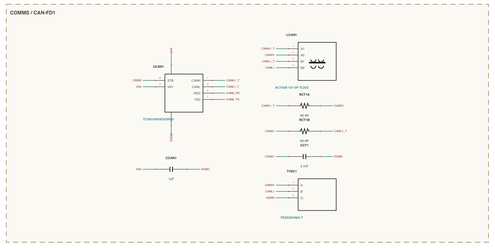
SBC / FS26



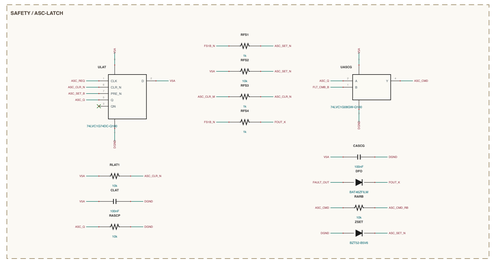
SBC / WAKE



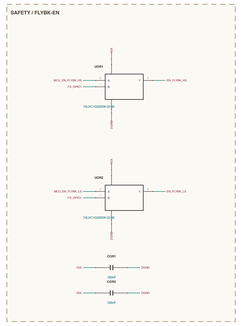
COMMS / CAN-FD1



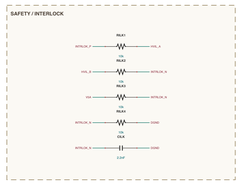
SAFETY / ASC-LATCH



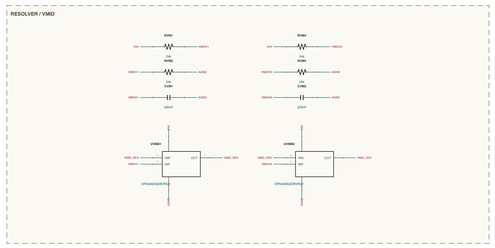
SAFETY / FLYBK-EN



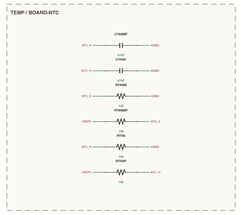
SAFETY / INTERLOCK



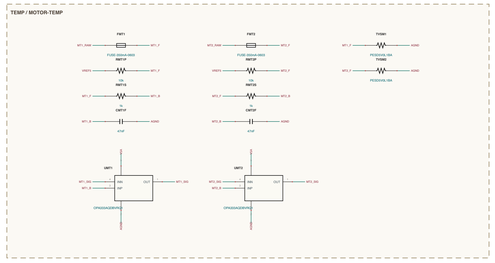
RESOLVER / VMID



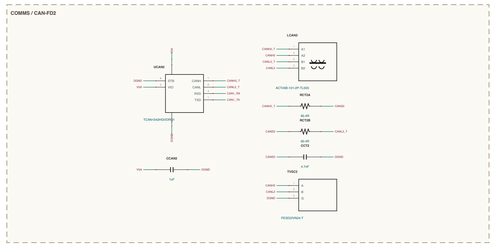
TEMP / BOARD-NTC



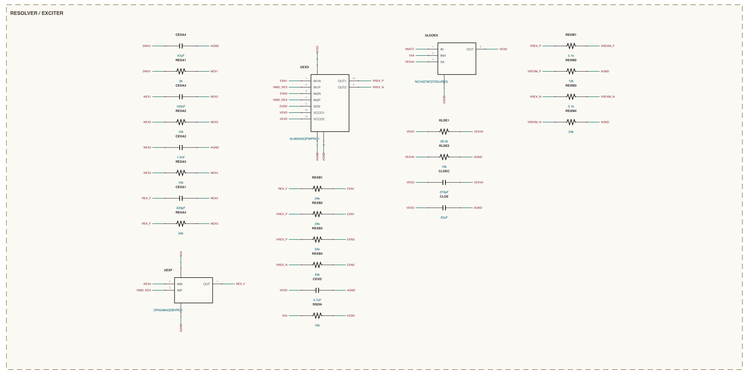
TEMP / MOTOR-TEMP



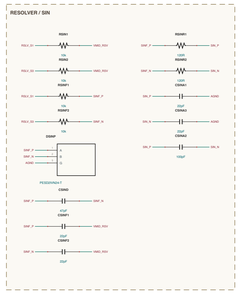
COMMS / CAN-FD2



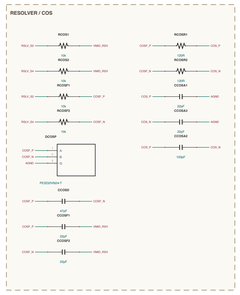
RESOLVER / EXCITER



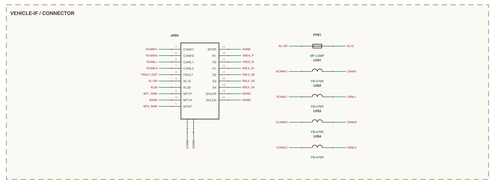
RESOLVER / SIN



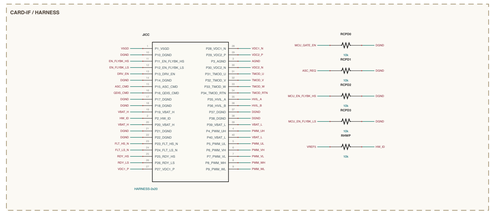
RESOLVER / COS



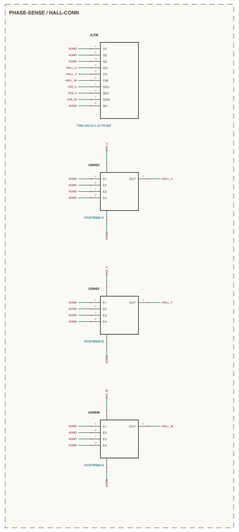
VEHICLE-IF / CONNECTOR



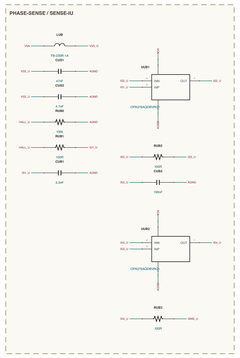
CARD-IF / HARNESS



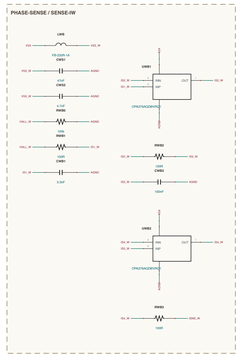
PHASE-SENSE / HALL-CONN



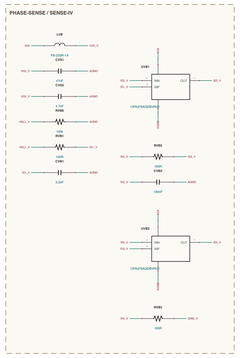
PHASE-SENSE / SENSE-IU



PHASE-SENSE / SENSE-IW



PHASE-SENSE / SENSE-IV



SHEET INDEX

220 kW pk Control card (LV) - 4 of 4

PHASE-SENSE - 4 sections

RESOLVER - 4 sections

SAFETY - 4 sections

SBC - 3 sections

TEMP - 3 sections

COMMS - 2 sections

CONTROL - 2 sections

VDC-RECEIVE - 2 sections

CARD-IF - 1 section

VEHICLE-IF - 1 section

rev A.11 - 26 sections - 288 components

NET NAMING

Internal junctions are named for what they join

Uxref+_vPIN: node at that IC pin e.g. U9VDC_VINP

Rxxtemp_W: midpoint of a series pair e.g. R8LA_W

Rxxtemp_vntc: tap between Rxxtemp+vntc e.g. RVDD_12

all others are explicit design nets

same name on another sheet = same net; joined ONLY at the

named connectors/solderfab (see INTERBOARD LINKS, design basis)

Traction Inverter — Control Card (S32K396 + FS26, ASIL D)

Traction Inverter 220 kW pk - Control card (LV), sheet 4 of 4

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